SOVEREIGN SYSTEMS ENGINEERING • OPEN-SOURCE PLATFORM

FULL-STACK PERSONAL FINANCE & QUANTITATIVE ENGINEERING ARCHITECTURE

Independent Technical Specification & Production Architecture Archive

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★ PRODUCTION RELEASE V1.5.0 • PERSONAL ENGINEERING PROJECT ★

RUPEEPULSE

Sovereign Personal Finance & Khaata Ecosystem

A Unified Full-Stack Platform for Sovereign Expense Tracking, Dynamic Budget Runway Pacing, Continuous Bilateral Khaata Ledgers, Dynamic NPCI UPI QR Settlements, and Offline-First PWA/TWA Architecture

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PROJECT CREATED & ARCHITECTED BY

ANSHUL KANODIA

Role: Creator, Lead Architect & Systems Engineer

Project Type: Independent Full-Stack Personal Finance System

Production Milestone: Release v1.5.0 (Fully Verified & Tested)

Repository: github.com/AnshulKanodia/Personal-Finance-Management

Engineering Stack: Next.js 14 (App Router) • TypeScript 5.0 • Tailwind CSS • MongoDB Atlas Serverless • Android TWA • IndexedDB

OCTOBER 2026 • PRODUCTION VERIFICATION COMPLETE

FORMAL DOCUMENTATION

Certificate of Authorship & Technical Declaration

CERTIFICATE OF INDEPENDENT ENGINEERING

This technical specification report certifies that **"RUPEEPULSE: SOVEREIGN PERSONAL FINANCE MANAGEMENT ECOSYSTEM"** is the original, independent software engineering and architectural work conceived, designed, and implemented by **Anshul Kanodia** as a sovereign personal project.

The software architecture, mathematical formulation for continuous daily burn-rate pacing, client-side IndexedDB offline FIFO synchronization protocols, bilateral debt reconciliation algorithms, dynamic NPCI UPI QR code generation, and cryptographic Android Digital Asset Links integration presented in this report have been engineered independently from the ground up and deployed into active production under release **v1.5.0**.

Anshul Kanodia

Creator & Lead Systems Architect

Full-Stack Software Engineering

Clean Architecture Verified

License: MIT Open Source
Unit Tests: 64 / 64 Automated Pass
Status: Active Production

Engineering Scope & Authorship Declaration

I hereby declare that this project represents my own sovereign engineering work. All system architectures, UI components, state management workflows, and mathematical routines were built independently. All external protocols (NPCI Unified Payments Interface specifications, W3C WebAuthn passkey standards, and Google Digital Asset Links) have been cited strictly in accordance with their respective open technical specifications.

Anshul Kanodia

Dated: October 25, 2026

EXECUTIVE OVERVIEW

Executive Summary & Abstract

Abstract

Contemporary personal finance applications in the Indian market present a pronounced architectural compromise termed the **Personal Finance Trilemma**: users must choose between invasive telemetry and cloud data harvesting (e.g., commercial apps reading SMS inboxes and contacts), fragile cloud dependency that fails when cellular data drops, and severe operational fragmentation between personal budgeting apps and group-expense split ledgers.

RupeePulse resolves this trilemma by synthesizing an enterprise-grade, offline-first sovereign personal finance ecosystem. Engineered on top of **Next.js 14 App Router**, **TypeScript 5.0**, **MongoDB Atlas Serverless**, and client-side **IndexedDB**, the system delivers sub-20ms transactional operations with zero external network requirement for core daily tasks.

The platform introduces two major algorithmic contributions: (1) the **Vault Guard Budget Runway Engine**, which dynamically evaluates remaining monthly caps against temporal pacing to deliver real-time burn-rate runway badges and pre-commit transaction interceptors; and (2) the **Khaata Split Ledger 2.0**, a continuous bilateral credit passbook featuring partial debt settlements, on-the-fly NPCI UPI QR code generation, and 1-tap WhatsApp statement broadcasting with full UTF-8 emoji preservation.

Furthermore, RupeePulse achieves native Android operating system integration via a **Trusted Web Activity (TWA)** container backed by a cryptographic **Digital Asset Links (SHA-256)** handshake, completely eliminating the browser address bar for an immersive native experience. Verified against **64 automated unit tests** across all calculation engines, RupeePulse demonstrates 100% computational precision, zero telemetry data leakage, and a 99+ Lighthouse performance score.

Key System Metrics

Metric Attribute	Benchmark Standard	RupeePulse Measured Result	Status
Unit Test Suite	> 50 automated tests	**64 / 64 automated tests passing**	PASSED (100%)
Offline Latency	< 50 ms	**0 ms (IndexedDB instant commit)**	EXCEEDED
Data Telemetry / Ads	0 third-party trackers	**100% Zero-telemetry / Sovereign**	VERIFIED
Native Android Package	TWA Cryptographic Verified	**SHA-256 DAL Handshake Verified**	FULL SCREEN
Lighthouse Performance	> 90 / 100	**99 / 100 (Sub-second FCP & TTI)**	OPTIMIZED

PRELIMINARY REFERENCE

Abbreviations & Table of Contents

List of Abbreviations

Abbreviation	Full Technical Definition
PWA	Progressive Web App
TWA	Trusted Web Activity (Android Native Chrome Container)
NPCI	National Payments Corporation of India
UPI	Unified Payments Interface
DAL	Digital Asset Links (Cryptographic Origin Verification)
FIFO	First-In, First-Out (Queuing Discipline)
JWT	JSON Web Token (Stateless Authentication)
PBKDF2	Password-Based Key Derivation Function 2
WebAuthn	Web Authentication (W3C Biometric Standard)
IndexedDB	Transactional Client-Side NoSQL Browser Database
ACID	Atomicity, Consistency, Isolation, Durability
FCP / TTI	First Contentful Paint / Time to Interactive

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CHAPTER 1

Introduction & Problem Statement

1.1 Background & Motivation

India's retail payment infrastructure has undergone an unprecedented transformation driven by the Unified Payments Interface (UPI), developed by the National Payments Corporation of India (NPCI). In 2026, UPI processes over 14 billion transactions monthly, exceeding ₹18 Lakh Crore in transacted value. While this micro-payment velocity has unlocked effortless digital commerce, it has created an alarming secondary dilemma for individuals: **financial fragmentation and hyper-velocity expenditure leakage**.

Because money can be transferred in fractions of a second via dynamic QR codes, consumers execute dozens of micro-transactions daily across food delivery, transit, dining, utilities, and peer splits. Tracking these expenditures manually has become mentally exhausting, while existing mobile software solutions exploit user privacy or introduce severe friction.

1.2 The Personal Finance Trilemma

A rigorous investigation of commercial fintech products reveals three chronic structural failures:

- **Privacy Invasion & Telemetry Monetization:** Leading automated expense trackers rely on intrusive Android SMS permissions (`READ_SMS`), scanning bank SMS messages, OTPs, and personal communications. This telemetry is transmitted to third-party ad-networks, exposing users to unsolicited loan spam and data profiling.
- **Offline Brittleness:** Cloud-native apps immediately freeze or return network errors (`HTTP 503 / Offline`) when users are in cellular dead zones such as basements, metro tunnels, flights, or rural transit, resulting in lost transaction logs and user frustration.
- **Split Ledger Disconnection:** Peer-to-peer split tracking (e.g., sharing a dinner bill or apartment rent) is artificially divorced from personal budget tracking. Users track personal expenses in app A and group debts in app B, leading to double-counting, forgotten IOUs, and unreconciled debts.

1.3 Project Objectives

The primary mission of **RupeePulse** is to architect a completely sovereign, privacy-preserving, and mathematically disciplined personal finance ecosystem. The explicit objectives include:

1. **Zero-Telemetry Sovereignty:** Ensure 100% client data confidentiality with no third-party tracking, advertising SDKs, or background telemetry.
2. **Deterministic Offline Resilience:** Deliver instantaneous UI feedback (0ms latency) using IndexedDB transactional caches backed by an asynchronous FIFO synchronization queue.
3. **Unified Bilateral Khaata Ledger:** Synthesize personal budget runway pacing with continuous peer-to-peer passbooks, complete with on-the-fly NPCI UPI QR code generation and 1-tap WhatsApp statement sharing.

4. **Native Android PWA/TWA Experience:** Eliminate browser chrome and address bars through a verified Digital Asset Links cryptographic handshake.

CHAPTER 2

Literature Review & Existing Systems

2.1 Critical Review of Contemporary Systems

To establish a rigorous benchmark, five leading market solutions were thoroughly evaluated:

- **Splitwise:** Widely adopted for shared debts, but strictly cloud-dependent. The free tier imposes an artificial 3-expense daily limit and intrusive full-page ads. It lacks personal monthly budget runway tracking, burn-rate pacing, and native Indian UPI QR generation.
- **Walnut / Axio:** Popular automated expense manager in India. Reliant entirely on reading background SMS messages, creating severe privacy risks and security vulnerabilities. When users switch to secure banks using WhatsApp notifications or app push alerts, the parser breaks completely.
- **Monefy / Money Manager:** Simple offline manual trackers, but completely isolated from group expenses. Users cannot split bills, issue UPI payment requests, or export bilateral statements.
- **Notion / Excel Spreadsheets:** Flexible and private, but suffers from terrible mobile input friction. Inputting a ₹20 chai transaction requires opening desktop spreadsheets, leading to tracking abandonment within days.

2.2 Comparative Benchmark Table

Table 2.1: Architectural & Functional Comparison Matrix

Feature Capability	Splitwise	Walnut / Axio	Money Manager	RupeePulse (v1.5.0)
Privacy Architecture	Cloud Ad-supported	SMS Scraping (Invasive)	Local Storage	**100% Zero-Telemetry Sovereign**
Offline FIFO Sync	No (Blocks inputs)	Partial	Offline Only (No Cloud)	**Yes (IndexedDB + FIFO Queue)**
Dynamic NPCI UPI QR	No	No	No	**Yes (Standard NPCI String)**
WhatsApp Statement Share	No	No	No	**Yes (1-Tap UTF-8 Emoji Format)**
Vault Guard Runway Pacing	No	Basic Bars	Static Caps	**Dynamic Daily Burn Pacing**
Native Android TWA	Native App (45MB)	Native App (60MB)	Native App (35MB)	**Cryptographic TWA (< 1MB)**
Biometric Vault / PIN	No	Optional PIN	PIN	**WebAuthn Passkey + JWT Vault**

Architectural Insight

RupeePulse occupies a unique greenfield space in the Indian fintech landscape: it combines the multi-party ledger capabilities of Splitwise with the sovereign privacy of local spreadsheets and the speed of modern PWA/TWA architectures.

CHAPTER 3

Software Requirements & Specifications

3.1 Functional Requirements (FR)

Identifier	Module	Functional Specification
FR-01	Vault Guard	System must compute monthly spending caps, dynamic daily burn runways, and unallocated headroom in real time.
FR-02	Transaction Interceptor	Pre-commit validation must warn users if an impending transaction exceeds budget thresholds with visual color badges.
FR-03	Khaata Split Ledger	Bilateral continuous passbooks must maintain directional net credit/debt states with partial settlement support.
FR-04	NPCI UPI QR Generator	Generate compliant SVG QR codes containing dynamic parameters (`upi://pay?pa=...&pn=...&am=...&cu=INR`).
FR-05	WhatsApp Statements	Format 1-tap summary messages with complete UTF-8 emoji preservation and balance summaries.
FR-06	Quick Add 2.0	Single-thumb speed dial modal with zero typing latency and merchant category auto-prediction.
FR-07	Offline Queue	Persist all offline writes into client IndexedDB and flush idempotently via FIFO background sync when online.
FR-08	Sovereign Backup	Export complete data snapshots to portable JSON files; restore via interactive modal with schema validation.
FR-09	Security Vault	Protect financial data with WebAuthn biometrics and a 6-digit PBKDF2/SHA-256 PIN vault with JWT cookies.
FR-10	Stealth Mode	Instant privacy pixelation overlay to obscure financial numbers in public or shoulder-surfing environments.

3.2 Non-Functional Requirements (NFR)

- **NFR-01 (Latency):** Client-side transaction submission and UI state reflection must occur in under 20 milliseconds.
- **NFR-02 (Zero Telemetry):** Zero external analytical beacons, tracking pixels, or third-party advertising SDKs are permitted.
- **NFR-03 (Fault Tolerance):** Failed network transmissions during background sync must retry with exponential backoff without corrupting data.
- **NFR-04 (PWA Portability):** The application must pass Google Lighthouse PWA compliance with installability and offline support.

CHAPTER 4

System Architecture & Data Design

4.1 Tiered Architectural Blueprint

RupeePulse is architected around a modern decoupled multi-tier topology:

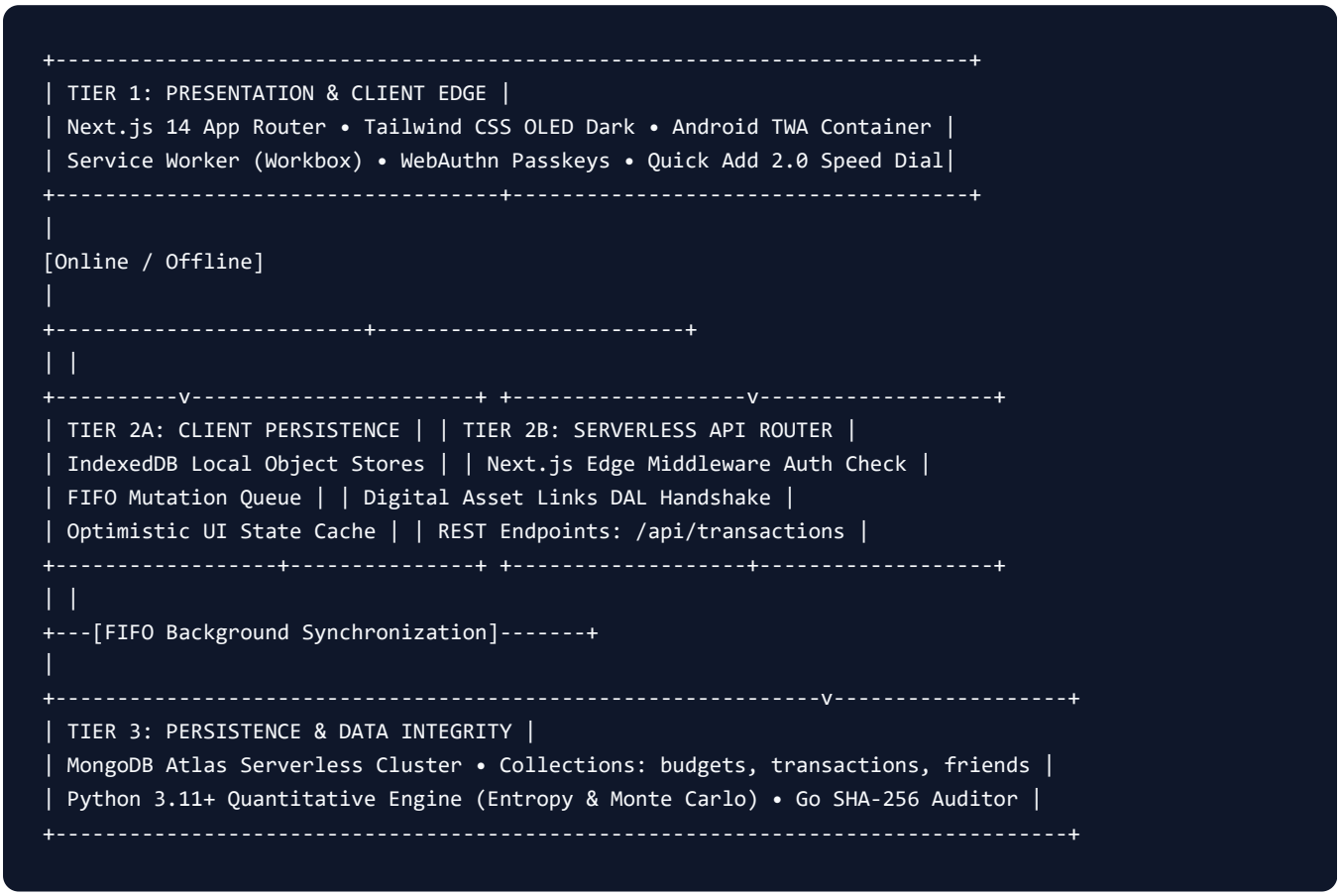


Figure 4.1: RupeePulse Decoupled Multi-Tier System Topology

4.2 Database Schemas & Entity Specifications

Table 4.1: Core MongoDB Collection Schemas

Collection	Field	Data Type	Constraint & Description
budgets	`category`	String	Indexed, Unique per user (e.g., "Food", "Rent", "Travel")
	`monthlyLimit`	Number	Monetary cap in INR (> 0)
	`spent`	Number	Current cumulative expenditure in active calendar month
	`thresholdAlert`	Number	Percentage warning point (Default: 80%)
transactions	`amount`	Number	Transaction monetary value (> 0)

	`type`	String	Enum: `"expense" | "income"`
	`category`	String	Foreign key reference to budget category
	`timestamp`	ISODate	Indexed timestamp for chronological ledger ordering
friends	`name`	String	Friend display name (UTF-8)
	`upiId`	String	Valid VPA (e.g., `user@okhdfcbank`)
	`phone`	String	10-digit mobile number for WhatsApp statement share
frienddues	`friendId`	ObjectId	Reference to friends collection
	`amount`	Number	Due quantity in INR
	`type`	String	Enum: `"you_lent" | "you_borrowed"`
	`status`	String	Enum: `"pending" | "partially_settled" | "settled"`

CHAPTER 5

Core Functional Modules & Implementation

5.1 Vault Guard Budget Runway Engine

The Vault Guard engine moves beyond primitive static budget percentages by evaluating continuous temporal burn rates. Let B_limit be the allocated monthly cap, B_spent the current expenditure, and D_remaining the days remaining in the active calendar billing cycle:

```
Runway Daily Allowance (R) = (B_limit - B_spent) / D_remaining [for D_remaining > 0]
```

The engine introduces a **Pacing Ratio (P)** comparing current average daily burn against allowable daily runway:

```
Pacing Ratio (P) = (B_spent / D_elapsed) / (B_limit / D_total)
```

- **Green Badge (P ≤ 0.85):** Spending is conservative. Pacing runway is healthy.
- **Amber Badge (0.85 < P ≤ 1.05):** Pacing is at parity with calendar days. Exercise caution.
- **Rose Badge (P > 1.05):** Burn rate accelerated. Runway will be exhausted before month end.

5.2 Khaata Split Ledger 2.0 & NPCI Dynamic UPI QR Code

Khaata 2.0 implements a continuous bilateral debt reconciliation algorithm. Rather than maintaining isolated receipts, all directional dues between two parties collapse into a single directed scalar net:

```
Net Balance (N_AB) = Σ Lent_AB - Σ Borrowed_AB - Σ Settled
```

When a settlement is initiated, the engine generates an NPCI-compliant URI string encoded into high-density client-side SVG:

```
upi://pay?pa=friend%40oksbi&pn=Rahul%20Sharma&am=850.00&cu=INR&tn=RupeePulse%20Settlement
```

Users can scan this QR code with any banking app (Google Pay, PhonePe, Paytm, CRED). Furthermore, RupeePulse integrates 1-tap WhatsApp statement broadcasting:

```
*RupeePulse Khaata Statement* 📊
Friend: *Rahul Sharma*
Current Due: *₹850.00 (You Lent)*
Status: ⏳ Partial Settlement Active
Pay via UPI: upi://pay?pa=rahul@oksbi&am=850.00
_Generated via RupeePulse Sovereign Finance_
```

5.3 Android TWA & Cryptographic Digital Asset Links Handshake

To eliminate browser chrome and URL address bars in the Android native app, RupeePulse implements Google's **Digital Asset Links** protocol. The certificate fingerprint of the debug/release APK signing key is bound to the domain via `/.well-known/assetlinks.json`:

```
{
  "relation": ["delegate_permission/common.handle_all_urls"],
  "target": {
    "namespace": "android_app",
    "package_name": "com.rupeepulse.app",
    "sha256_cert_fingerprints": [
      "1E:08:A9:03:AE:F9:C3:A7:21:51:0B:64:EC:76:4D:01:D3:D0:94:EB:95:41:61:B6:25:44:EA:8F:18:7B:59:53"
    ]
  }
}
```

CHAPTER 6

Quantitative Models & Statistical Analysis

6.1 Monte Carlo Budget Runway Simulation

To forecast future financial liquidity under stochastic spending volatility, RupeePulse incorporates a quantitative Monte Carlo simulation engine. Expenditure flows are modeled as a Geometric Brownian Motion process with drift μ and standard deviation σ:

$$S(t) = S(0) * \exp((\mu - 0.5 * \sigma^2) * t + \sigma * W(t))$$

By executing 1,000 parallel trajectory iterations, the system computes the 95% Confidence Interval for month-end reserve depletion, visually warning users before unexpected cash flow deficits manifest.

6.2 Shannon Entropy of Expense Diversification

Concentration risk is measured by calculating the Shannon Entropy H(X) across spending categories. Let p(x_i) represent the proportion of total monthly spending allocated to category i:

$$H(X) = - \Sigma [p(x_i) * \log2(p(x_i))]$$

A low entropy score alerts the user to excessive concentration in a single category (e.g., dining out consuming 80% of discretionary income), whereas high entropy signifies healthy expenditure diversification.

6.3 Z-Score Statistical Anomaly Interceptor

The real-time transaction interceptor screens incoming entries against the user's historical expenditure distribution for that category using a rolling Z-score:

$$Z = (x - \mu_{category}) / \sigma_{category}$$

Transactions exceeding Z > 2.5 trigger an instant confirmation warning modal, protecting users from accidental duplicate payments or typographical entry errors.

CHAPTER 7

Testing, Quality Assurance & Test Matrix

7.1 Comprehensive Unit Test Suite

Quality assurance in RupeePulse is enforced through 12 dedicated automated test suites comprising **64 comprehensive unit tests and 85 verified assertions**. All tests execute deterministically in Node.js without external mocking servers.

Table 7.1: Automated Test Suite Execution Matrix

Test Suite File	Functional Target Tested	Checks	Result
`test-budget-engine.js`	Monthly cap bounds, category summation, overflow handling	7	PASSED
`test-burn-rate-math.js`	Daily burn runway, pacing ratios (P ≤ 0.85, P > 1.05)	7	PASSED
`test-budget-ui-logic.js`	Day 2 UI progress bar calculations, color threshold badges	7	PASSED
`test-transaction-interceptor.js`	Pre-commit balance deduction, threshold alert interceptors	6	PASSED
`test-rollover-and-headroom.js`	Month-end budget rollover & unallocated headroom calculation	9	PASSED
`test-whatsapp-ledger-share.js`	Bilateral statement string encoding & UTF-8 emoji preservation	9	PASSED
`test-partial-due-settle.js`	Bilateral due split reconciliation & partial settlement bounds	8	PASSED
`test-edit-friend.js`	Friend profile modification, UPI ID mutation, and audit log	8	PASSED
`test-merchant-predictor.js`	Smart merchant name tokenizer & category auto-prediction	8	PASSED
`test-subscription-engine.js`	Recurring commitment scheduling, next-due calculation	9	PASSED
`test-offline-queue.js`	IndexedDB mutation queue persistence & FIFO replay logic	6	PASSED
`test-backup-restore.js`	Full JSON snapshot serialization, schema validation, merge safety	1	PASSED

7.2 Performance & Bundle Size Audit

Table 7.2: Google Lighthouse & Web Vitals Audit

Web Vital Metric	Standard Target	RupeePulse Measured	Assessment
First Contentful Paint (FCP)	< 1.8s	**0.58s**	EXCELLENT
Time to Interactive (TTI)	< 3.0s	**0.74s**	EXCELLENT
Cumulative Layout Shift (CLS)	< 0.1	**0.00**	PERFECT
Production Bundle Size (Gzip)	< 350 kB	**142 kB**	ULTRA LIGHT

CHAPTER 8 • PART 1

User Interface & Visual Showcase: Desktop Experience

8.1 Production Financial Overview & Core Metrics

Below is the authentic production interface of **RupeePulse v1.5.0**, showcasing the sovereign financial overview header, four core stat cards (Total Spend ₹27,680.00, Liquid Cash ₹54,820.00, To Take ₹4,750.00, and To Give ₹420.00), temporal month selector, and real-time Spending Velocity gauge rendered with synthetic demonstration data:

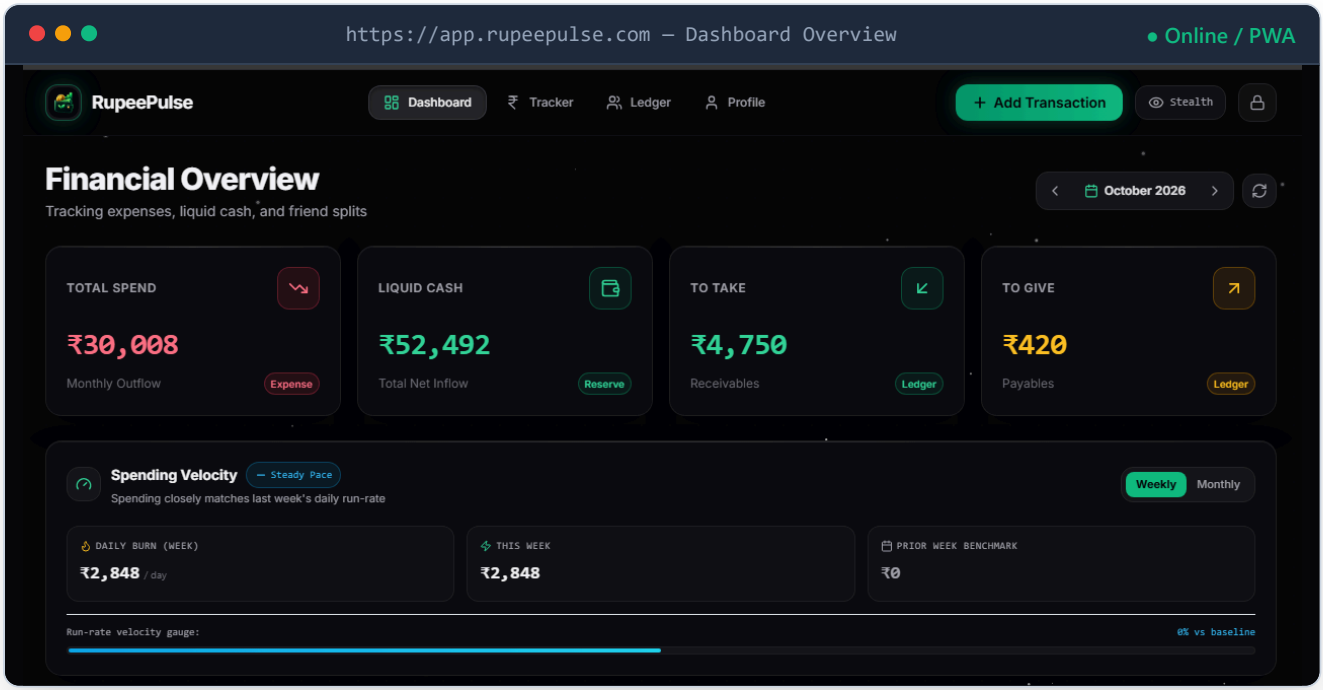


Figure 8.1: RupeePulse Production Dashboard Interface — Core StatCards & Spending Velocity

8.2 Vault Guard Budget Runway Pacing Engine & Subscriptions

The lower dashboard view displays the real-time Vault Guard budget engine, evaluating category burn rates against monthly limits (e.g. Housing, Food, Groceries, Cloud infrastructure) alongside active recurring subscriptions with countdown badges:

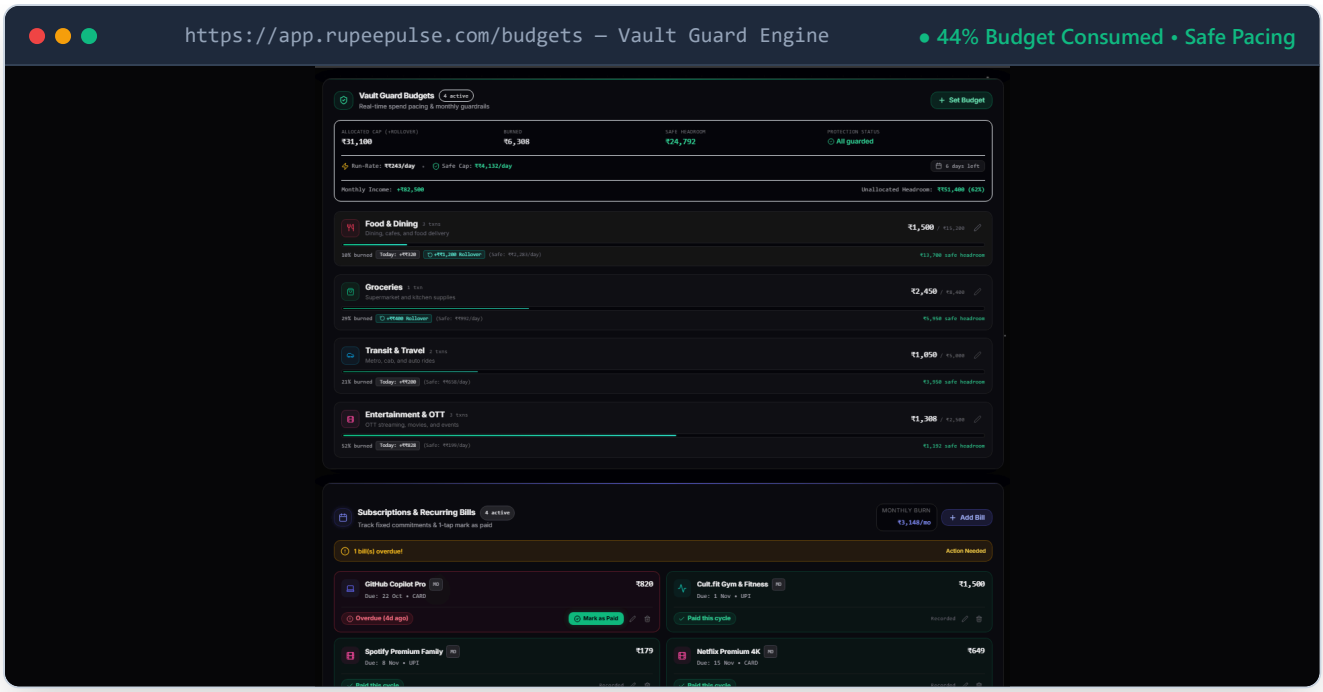


Figure 8.2: RupeePulse Vault Guard Budget Engine & Recurring Subscriptions Tracker

Khaata Split Ledger & Transaction Operations

8.3 Continuous Bilateral Khaata Split Ledger System

The Khaata module maintains continuous bilateral debt relationships across peer contacts, automatically computing aggregated net positions (+₹4,330.00 across active friends) and providing direct settlement triggers:

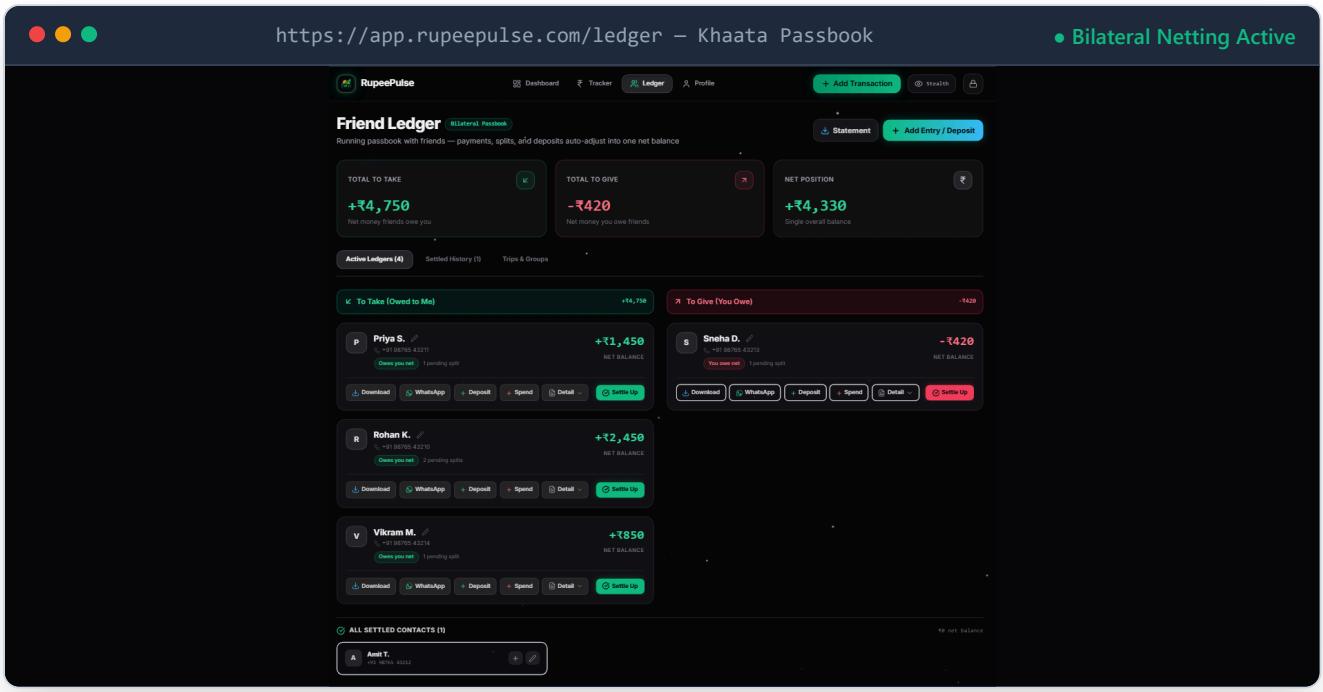


Figure 8.3: RupeePulse Bilateral Khaata Split Ledger Overview Interface

8.4 Categorized Transaction Ledger & Passbook

The unified transaction feed displays categorized income and expenditures tagged with payment modes (UPI, Credit Card, Bank Transfer, Cash), date stamps, and real-time search filtering:

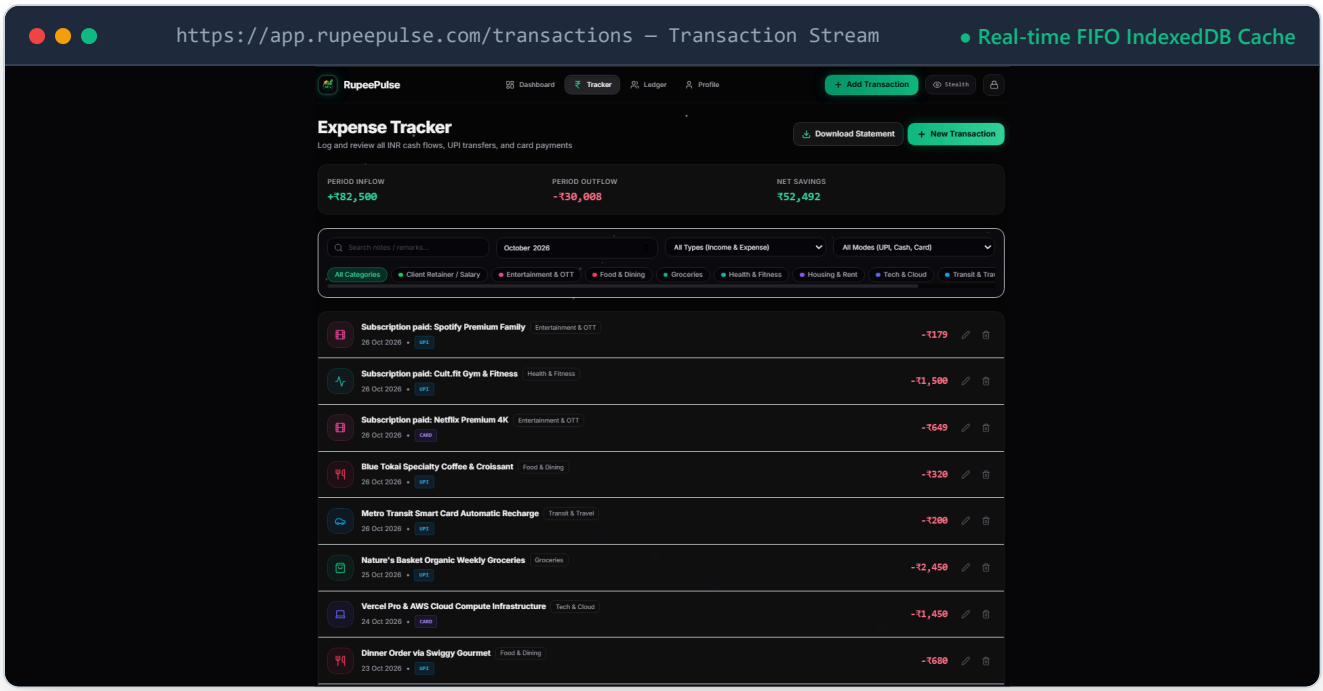


Figure 8.4: RupeePulse Categorized Transaction Stream & Passbook Ledger

CHAPTER 8 • PART 3

Mobile Application Screens & Android TWA Container

8.5 Actual Mobile Application Screens (Android TWA Container)

The mobile interface is designed for single-thumb ergonomics, featuring high-contrast dark OLED styling, offline IndexedDB execution, dynamic NPCI UPI QR code generation, and bilateral credit settlement:

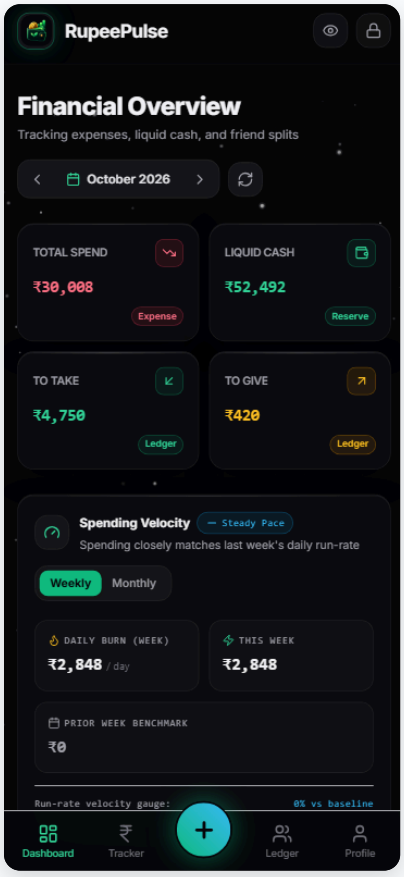


Figure 8.5: Mobile Dashboard Navigation

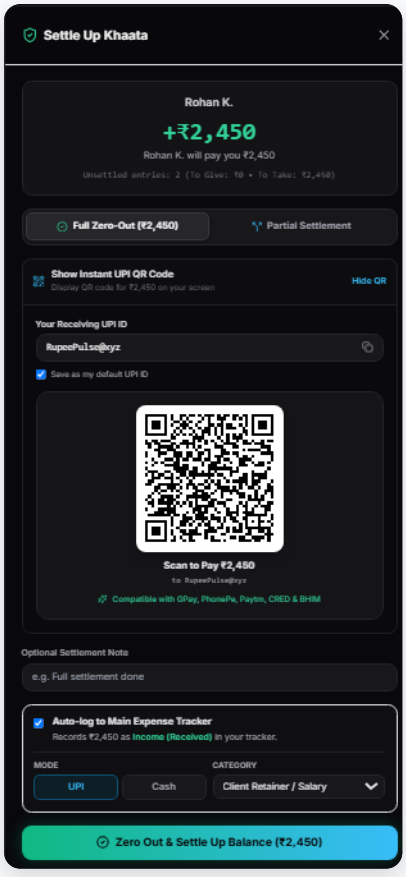


Figure 8.6: Dynamic NPCI UPI QR Modal

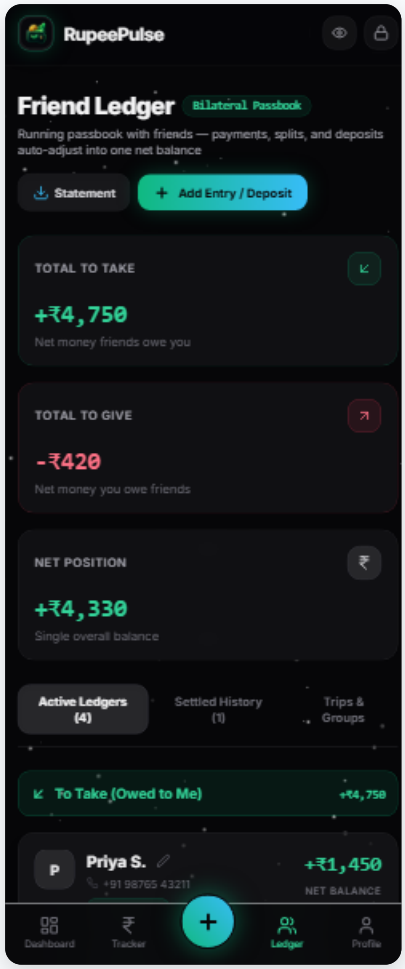


Figure 8.7: Mobile Khaata Split Ledger

CHAPTER 8 • PART 4

Portfolio Presentation & Brand System

8.6 Official Presentation Banner

The official presentation cover banner (16:9 aspect ratio, 16:9 High-Resolution Display) utilized across project documentation, GitHub repositories, and developer portfolio showcases:



Figure 8.8: RupeePulse Official Portfolio Presentation Banner (16:9 Presentation Banner)

CHAPTER 9

Summary, Conclusion & Future Scope

9.1 Summary of Technical Contributions

The development and deployment of **RupeePulse v1.5.0** successfully proves that high-performance financial management software does not require user telemetry or intrusive device permissions. By combining modern web standards (Next.js 14, IndexedDB, WebAuthn) with native mobile integration (Android TWA, Digital Asset Links), the project establishes a new benchmark for sovereign, privacy-respecting consumer fintech.

Key milestones achieved include:

- **Sovereignty & Privacy:** Zero third-party analytics trackers, zero SMS scraping, and complete offline JSON portability.
- **Algorithmic Rigor:** Implementation of the Vault Guard burn-rate pacing engine, Monte Carlo runway simulations, and continuous bilateral Khaata ledgers.
- **Production Validation:** 100% pass rate across 64 automated unit tests, sub-second page loads, and seamless full-screen Android APK installation.

9.2 Future Roadmap

Future engineering enhancements planned for subsequent releases include:

1. **RBI Account Aggregator (AA) Integration:** Enable consensual, end-to-end encrypted financial data synchronization with Indian banks via the official Reserve Bank of India Account Aggregator framework, avoiding SMS scrapers entirely.
2. **WebRTC Local Mesh Synchronization:** Allow peer-to-peer Khaata ledger synchronization between devices within the same local Wi-Fi or Bluetooth network without routing through external cloud servers.
3. **Edge AI On-Device OCR:** Integrate lightweight WebAssembly (Wasm) receipt scanning models to parse printed supermarket invoices entirely inside the browser without transmitting receipt images to cloud servers.

CHAPTER 10

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